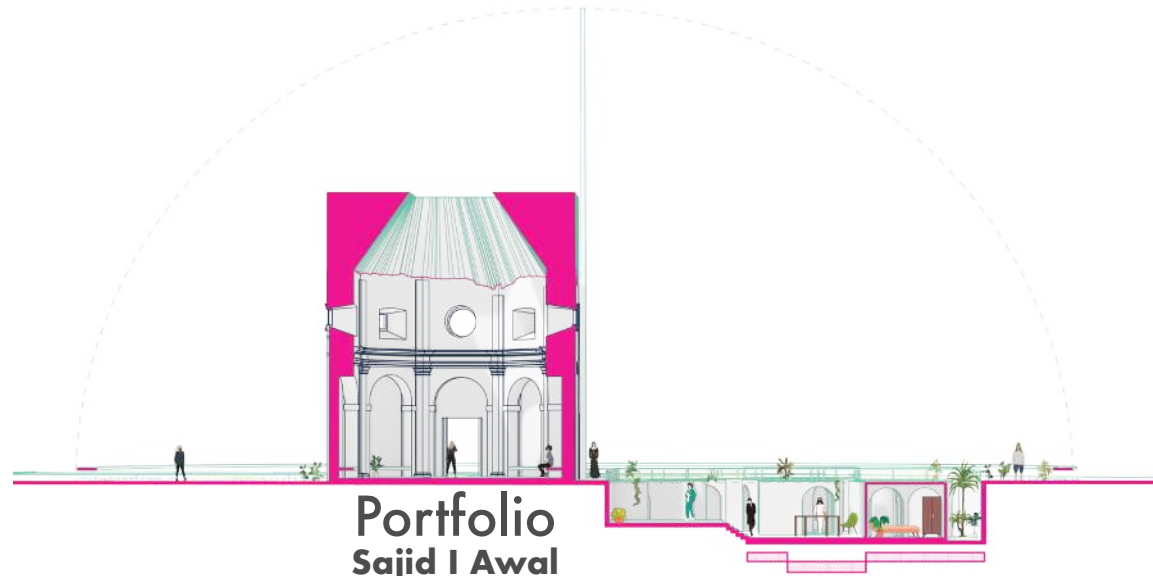
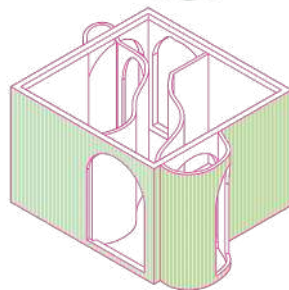
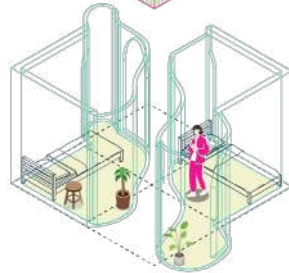
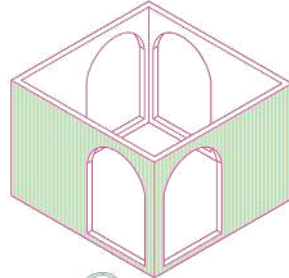
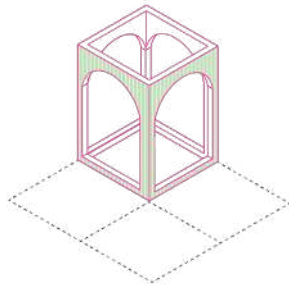




Portfolio
Sajid I Awal



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Pritila Waddedar Memorial Complex

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Raycasting3d in Grasshopper

PRITILATA WADDEGAR MEMORIAL COMPLEX

Third Year Project

Duration: 6 Weeks - 2013

Tools: AutoCAD, Photoshop, Illustrator, Maya

Pritilata Wadegdar was an revolutionary historical figure, who was the disciple of Masterda Surya Sen during the british colonial period in the indian sub-continent. The project was about designing a memorial complex on an imaginary site. The Concept was derived from a famous song of the famous poet who wrote the national song of Bangladesh, which can be translated as 'I will overcome the storm and won't fear it'. These concept is translated into design as overpowering ocean weave with overhead cantilever to show the mightiness and give the essence of revolution.

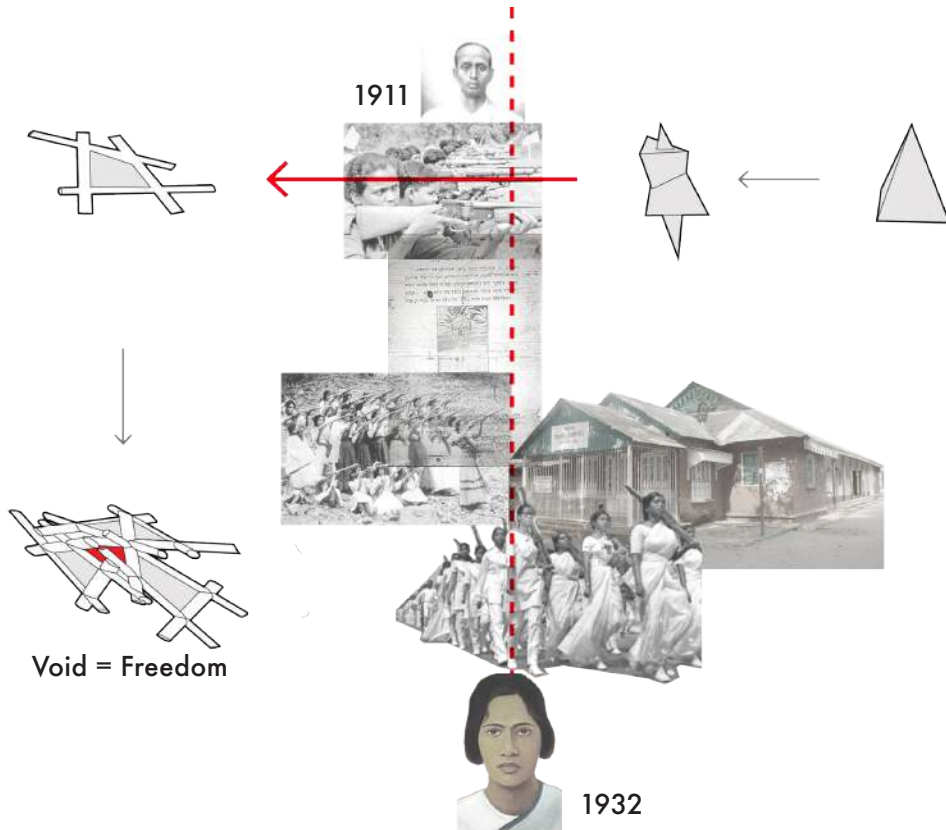
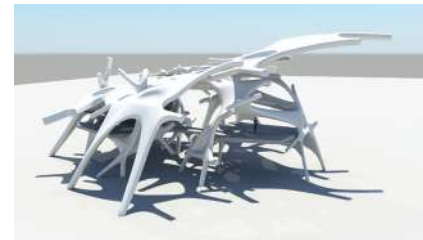
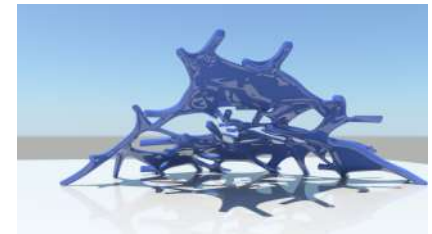
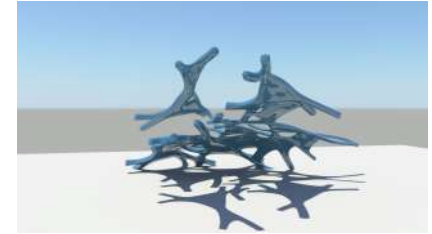
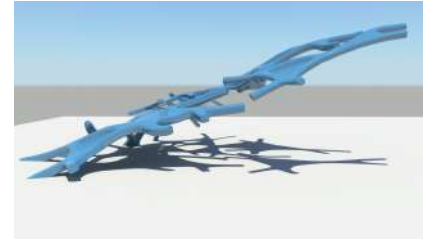
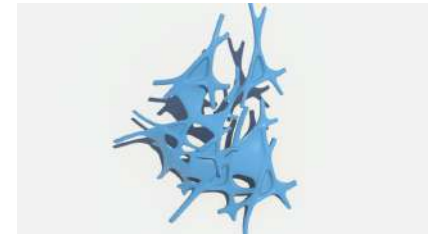




“ঝড় কে আমি করব মিতে, ডড়ব না তার ঝকুটিতে।”

তোমার খোলা হাওয়া
- রবিন্দ্রনাথ ঠাকুর

Ocean wave as Revolution





Handcrafted Model



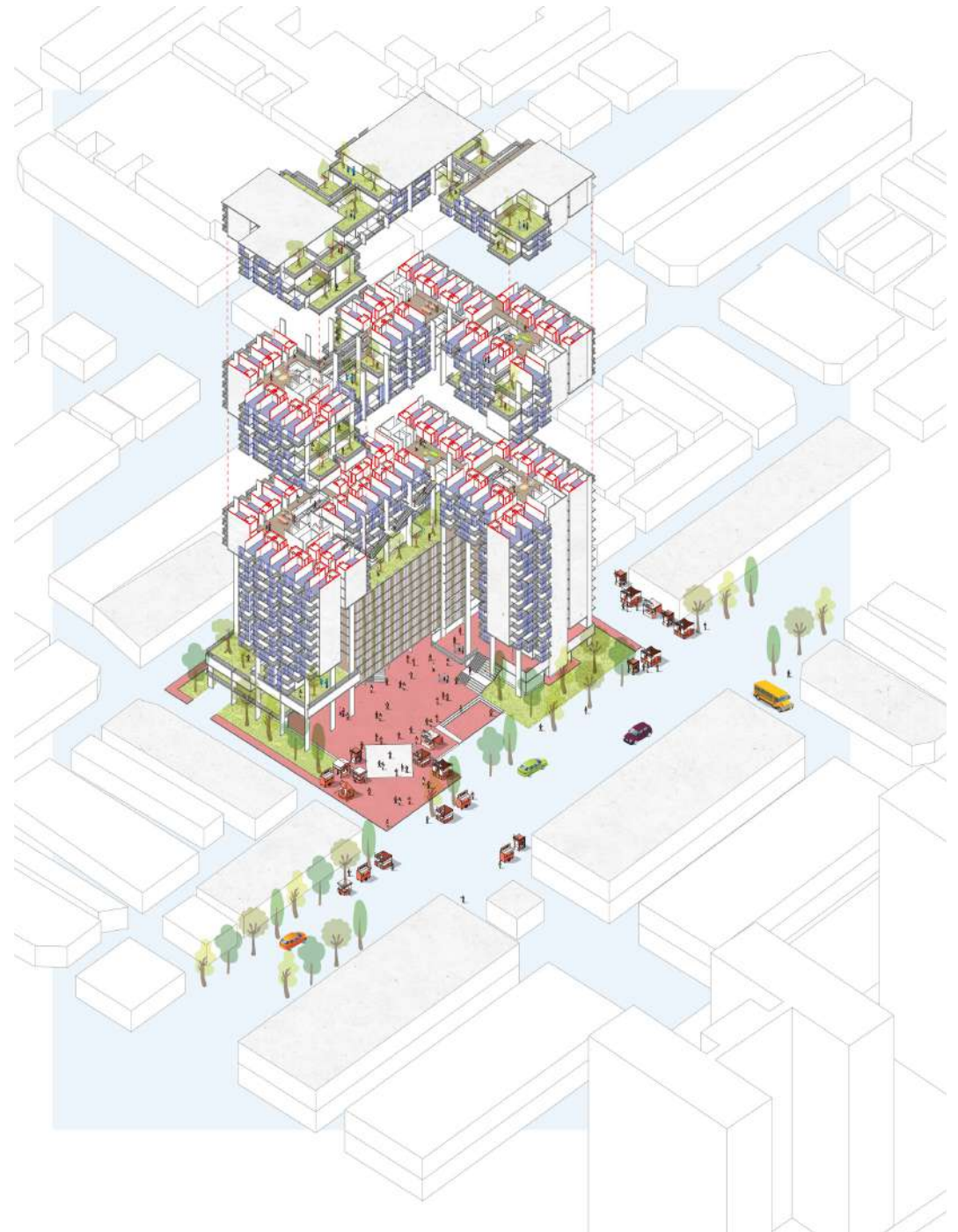
PUBLIC, PRIVATE & IN-BETWEEN SPACES IN STUDENT HOUSING

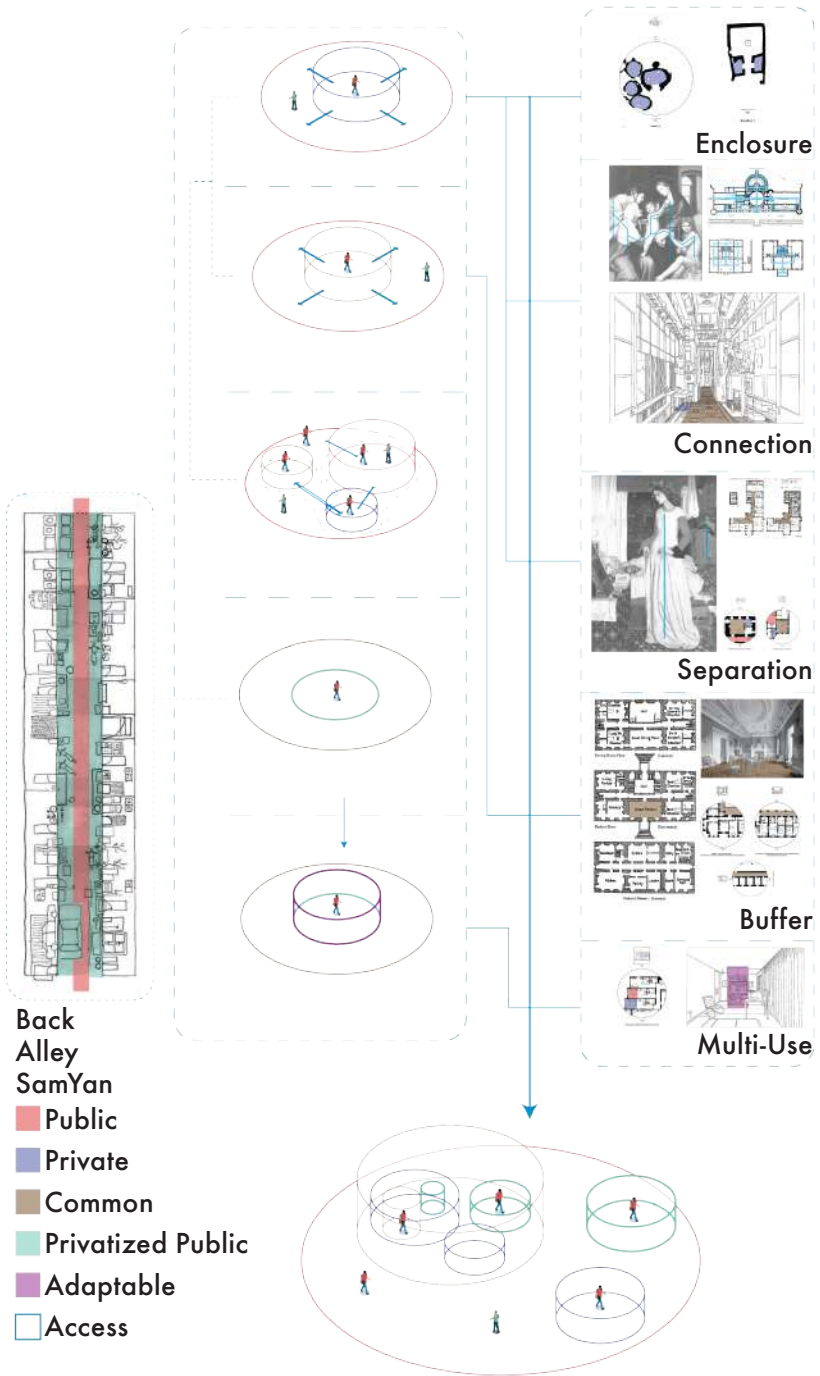
i+MARCH Thesis Project

Duration: 1 Year - (2020-2021)

Tools - AutoCAD, Rhino, DepthmapX, Photoshop, Illustrator

The right balance between public and private spaces is one of the most difficult issues in any housing design. Scholars argue that in student housing public and private spaces are equally important for promoting interaction as well as ensuring opportunities for self-development among the students. This research aims to establish definitions for public, private, and in-between spaces along with their relationships to achieve the right balance in the design of it which would eventually contribute not only to the students, but also the university and surrounding areas. Both qualitative and quantitative methods are used for this research and then the student housing is designed following the findings. This research reveals that public and private spaces coincide together in student housing. Whether a space is public or private can be identified based on four factors, which are accessibility, inclusiveness, visibility, and use of the space. Several in-between spaces like common, privatized public and adaptable spaces are also found following these derived factors which create the balance between publicness and privacy in space at certain periods of time. Accordingly, no such space can be termed as entirely private or public space and thus the balance cannot be determined in a quantitative way. Furthermore, the scale of the space or the way space is studied is also crucial in identifying the publicness or privacy of that space.





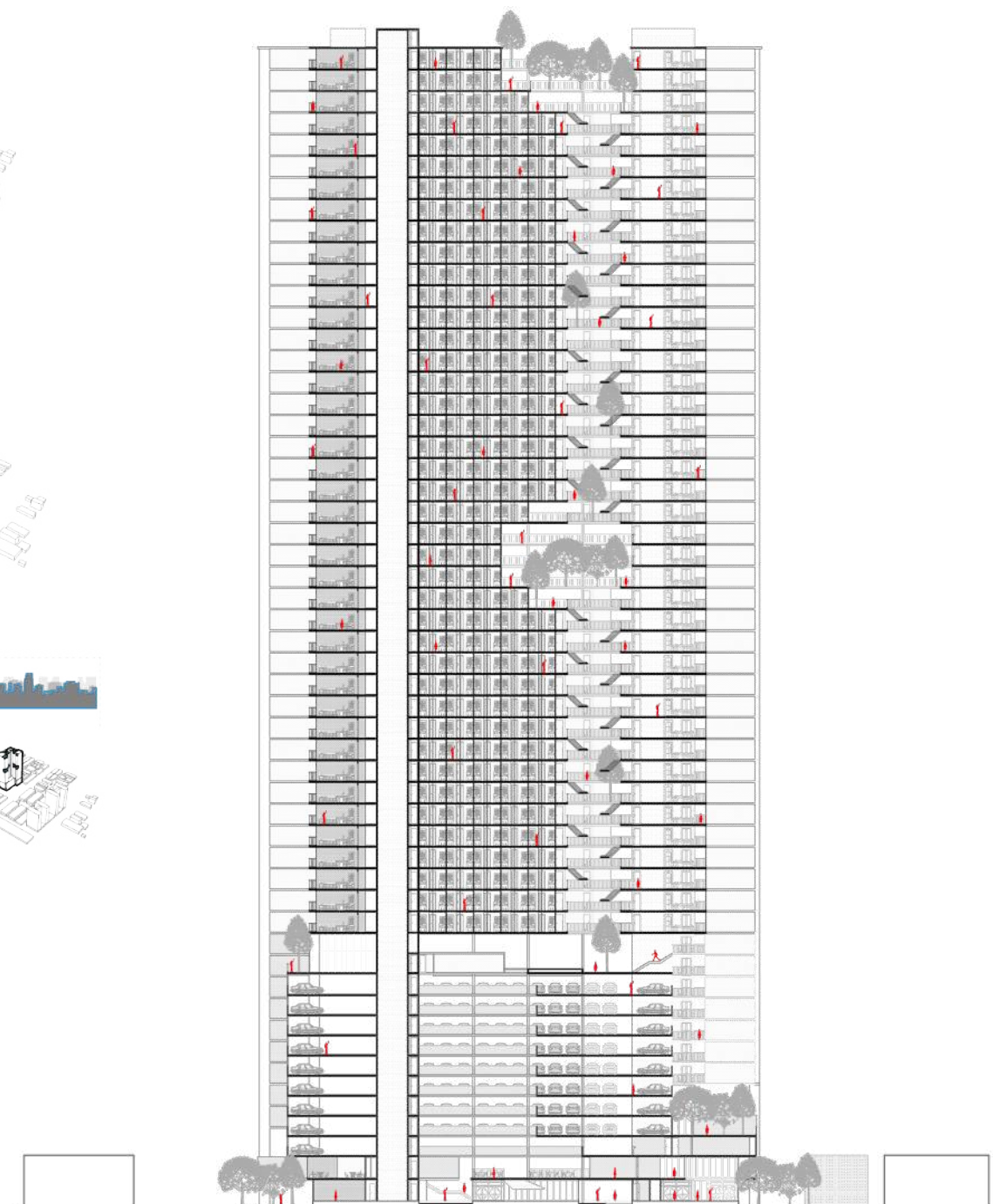
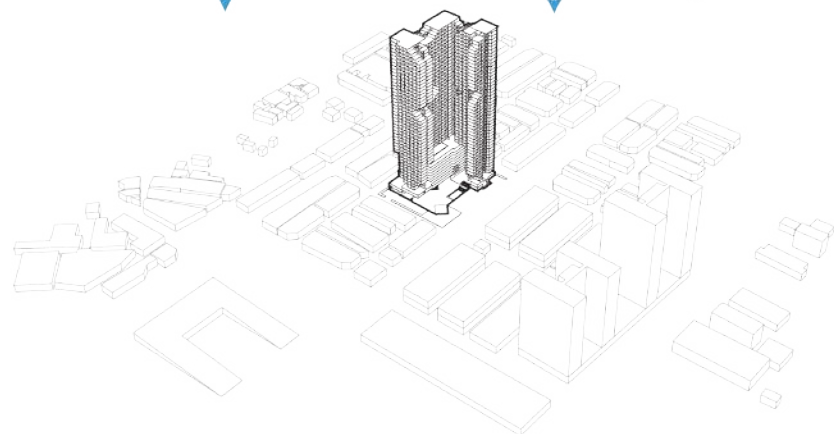
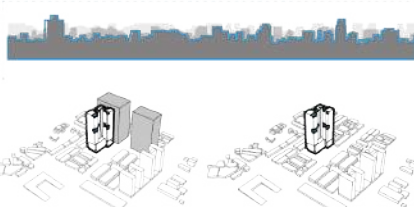
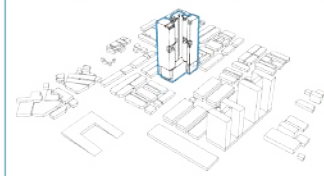
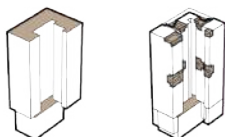
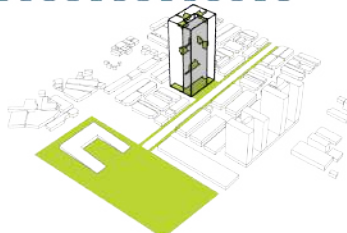
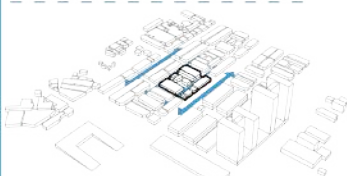
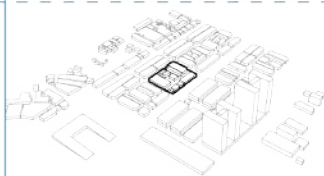
Context Study

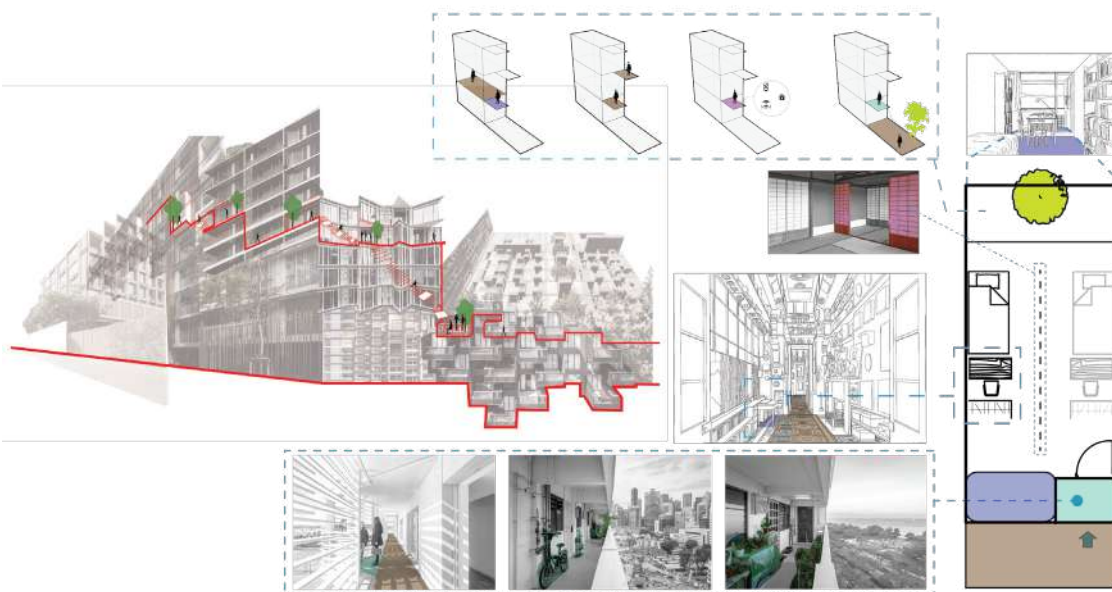
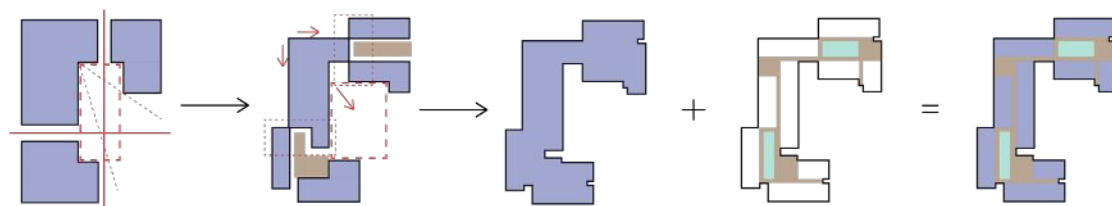


Research

Form Development

Context





Floor Plan Development



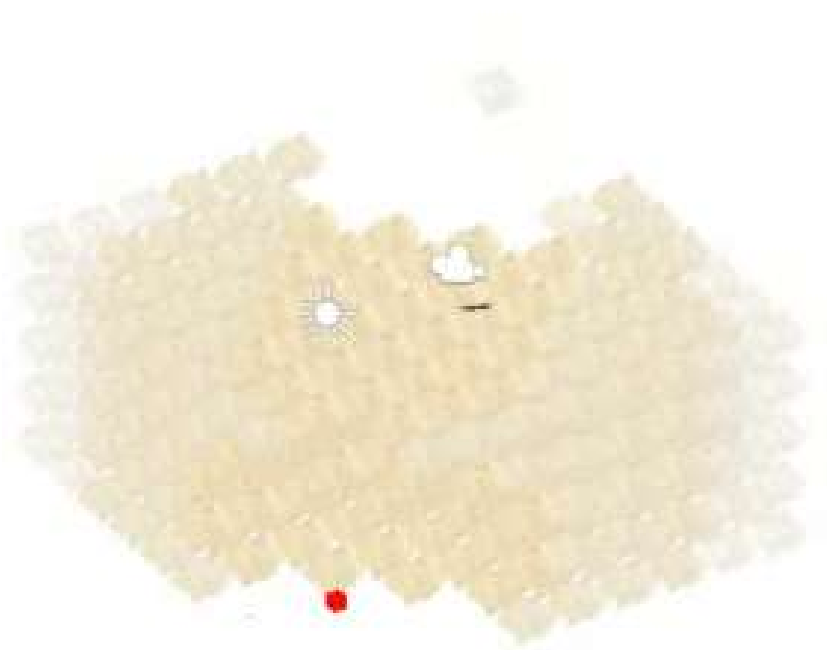
PRIVACY THROUGH VIRTUAL REALITY

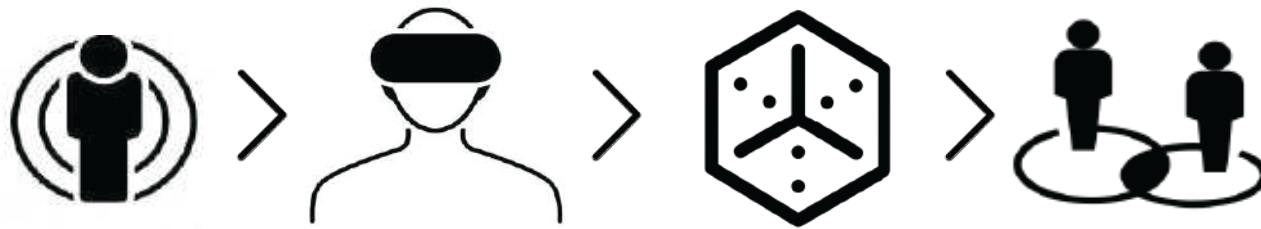
MSc Studio Project

Duration: 3 Months - (2022)

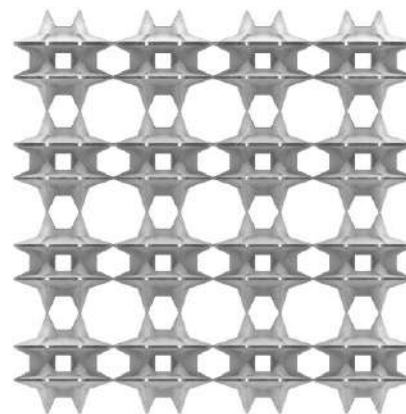
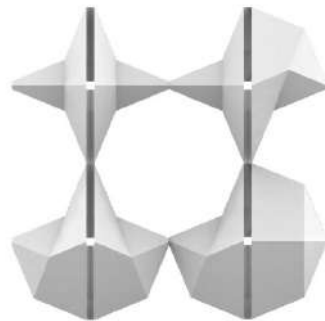
Tools - Grasshopper, Unity3D, HoloLens2

This is a studio project within a group of three, where we explored privacy of individual in public space with the use of VR. So, we created a virtual privacy shell to create boundary, which could be visible while using VR device. Hence, when needed visual privacy could be ensured. Here, we used MRTK toolkit plugin inside Unity3D to use Microsoft HoloLens2 to deploy our geometry. The geometry is derived from Grasshopper and imported into Unity3D. Field algorithm is used to create distance from the observer to the shell in real time.



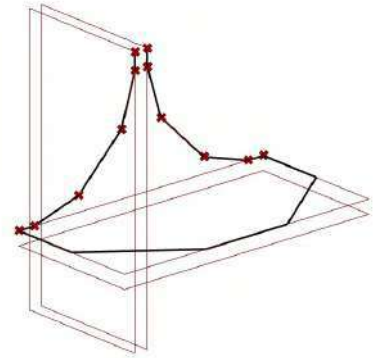
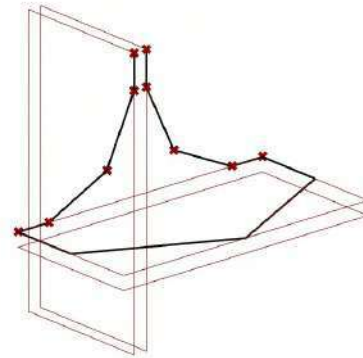
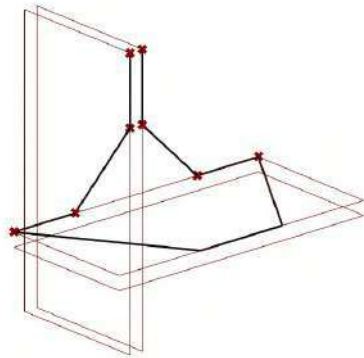
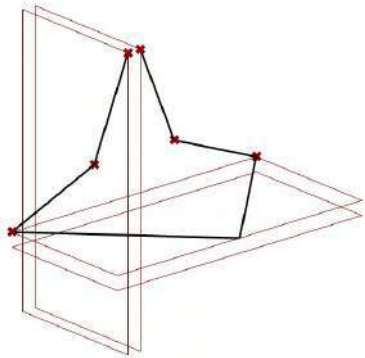


Concept Development



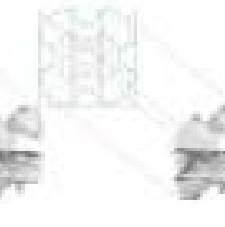
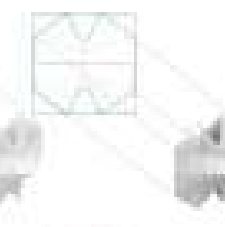
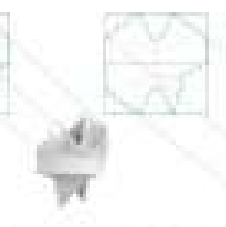
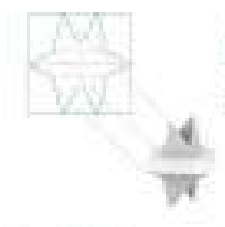
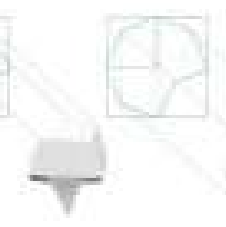
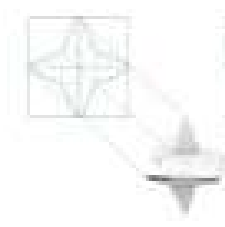
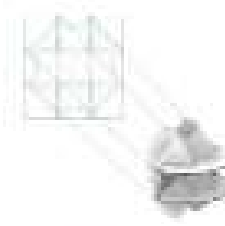
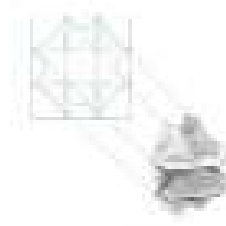
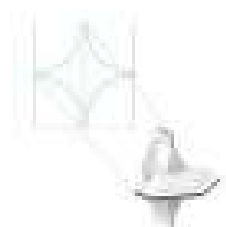
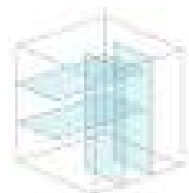
Form Development

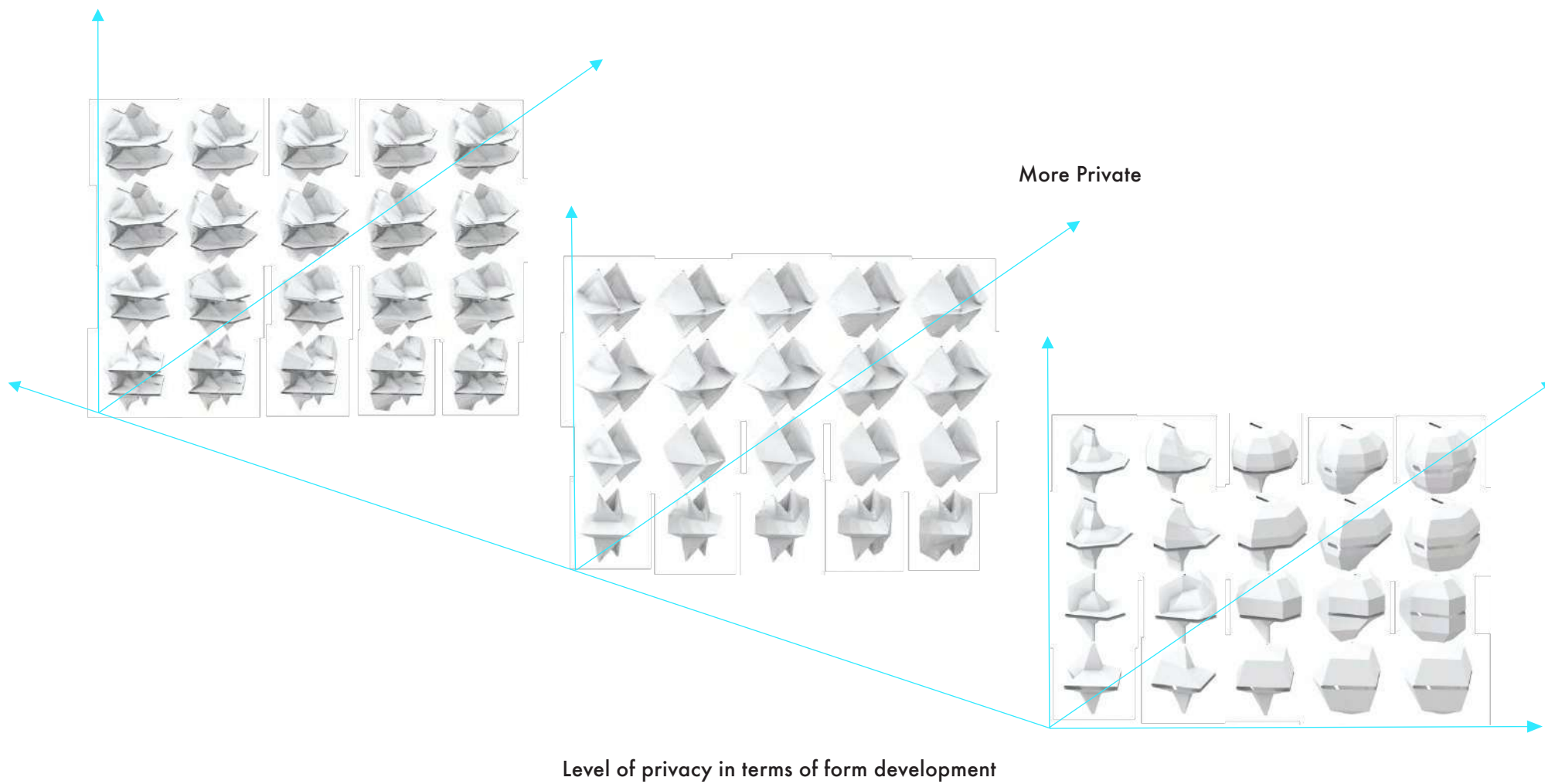

```
for (int i = 0; i < edges; i++)
{
    getMidpoints;
}
```

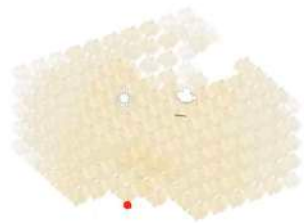
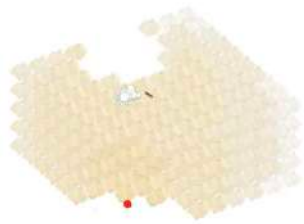
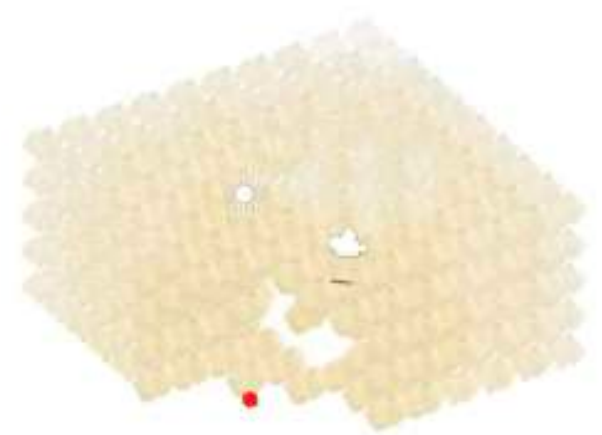


```
Transform myTransform = Transform.Rotation();
```

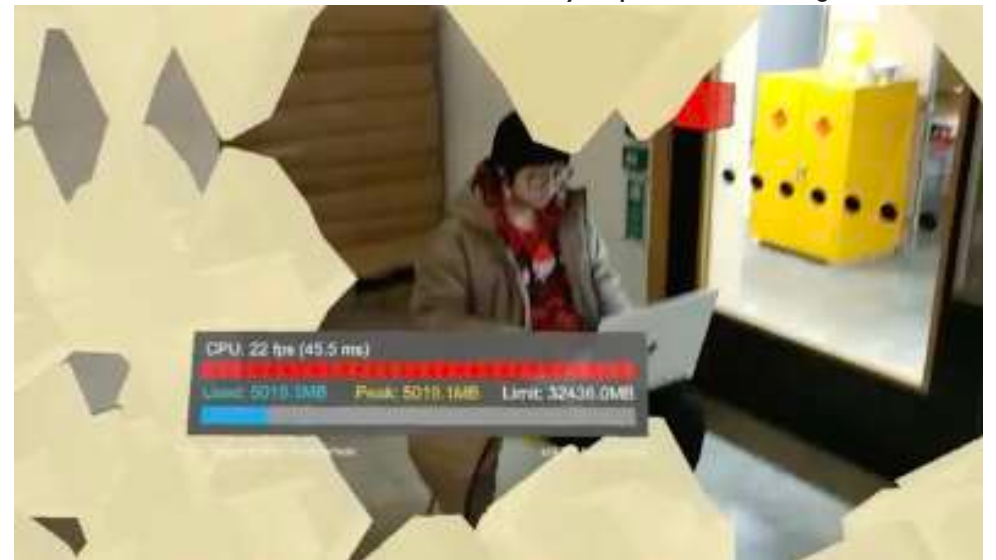








Privacy experienced through HoloLens2



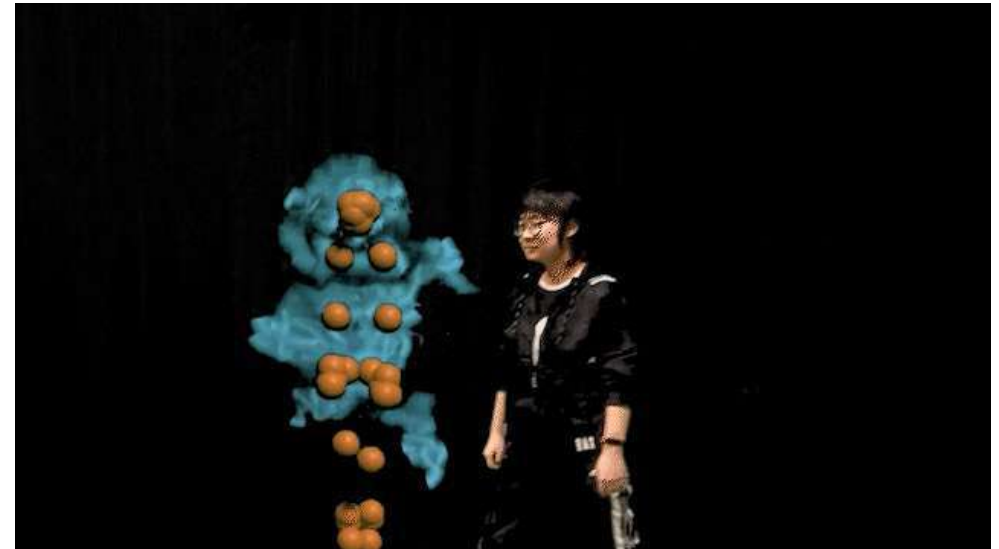
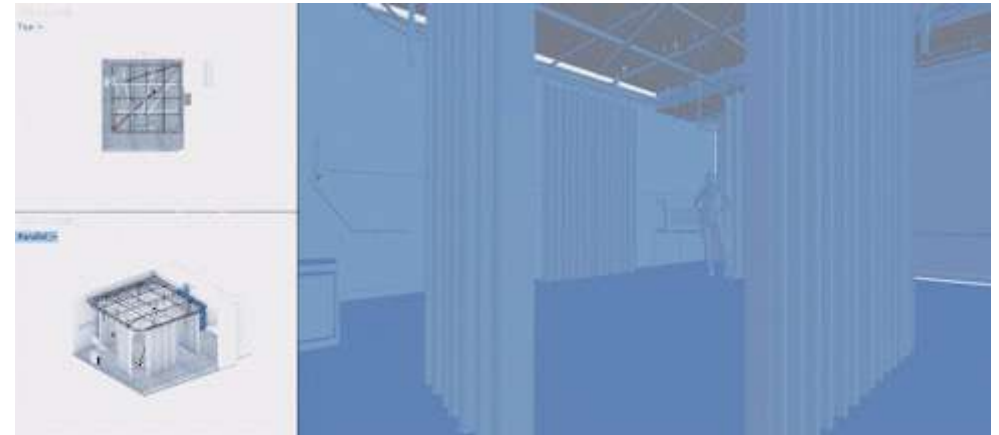
AUGMENTING PERFORMANCE In CYBER-PYSICAL SPACE

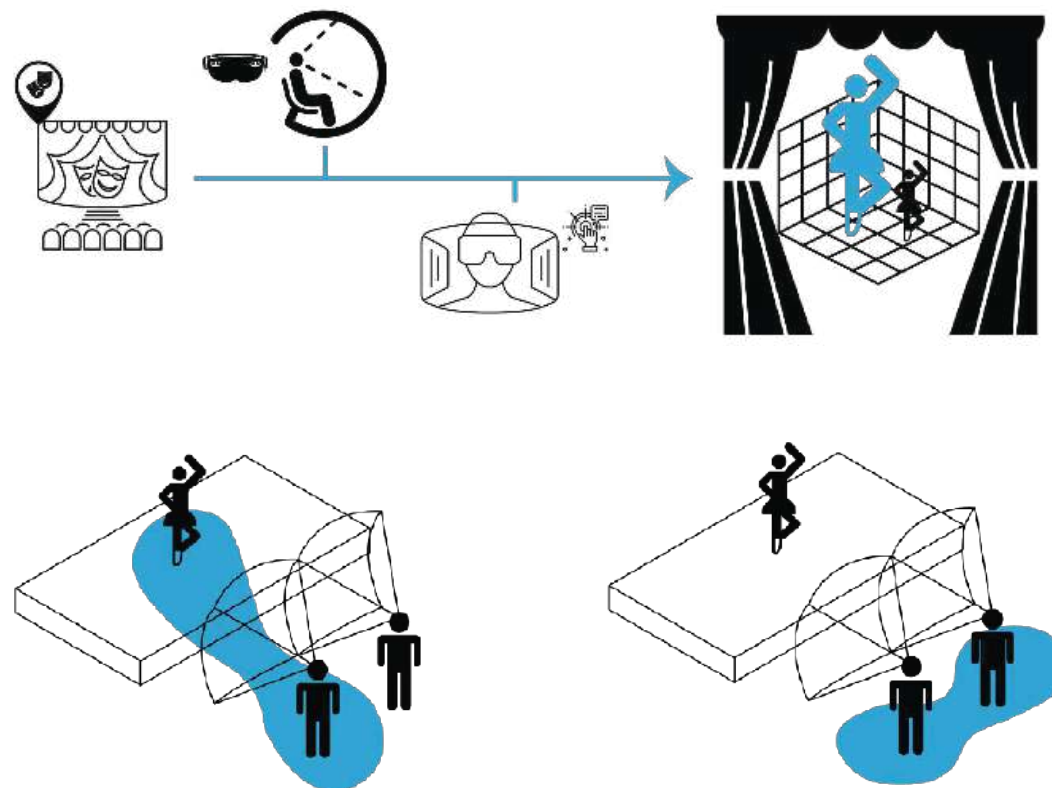
MSc Studio Project

Duration: 3 Months - (2023)

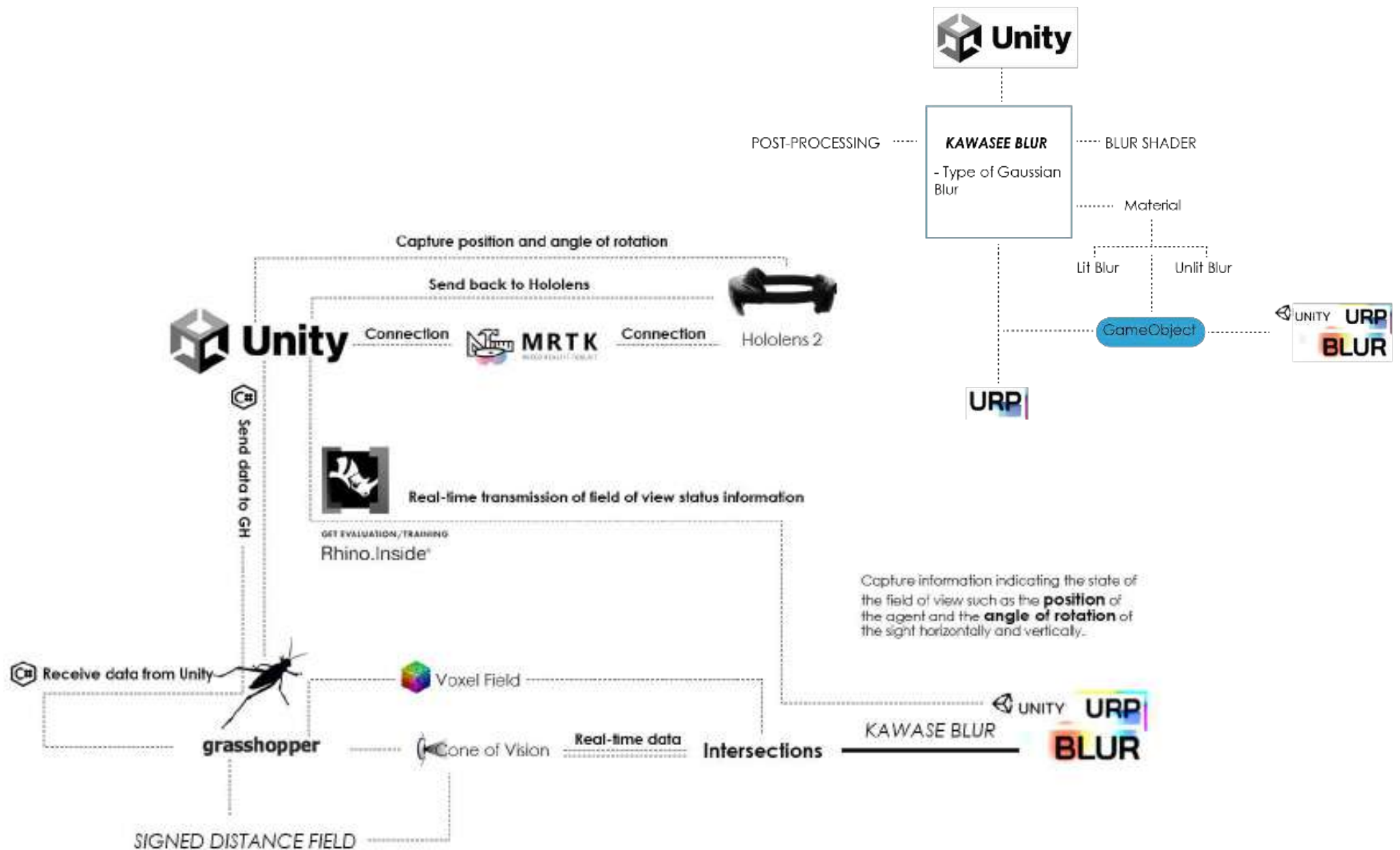
Tools - Grasshopper, Unity3D, HoloLens2

This work represents a continuation of an exploration of the metaverse undertaken collaboratively within a group of three. The project addresses a speculative, futuristic theatre context in which the audience can interact directly with the performer through an immersive experience. The investigation focused on two parallel aspects. First, we explored methods of introducing a de-liberate *blur* between multiple audience members to enhance immersion while preserving individual privacy. Second, we employed *Media Pipe Pose* to detect the audience's articulated body points, which were then translated into a Unity3D environment and ultimately deployed on HoloLens 2, allowing the experience to be perceived from the audience member's own perspective. *Agent-based modelling* was subsequently used to generate a dynamic mesh around these articulated points, enabling real-time responsiveness. This system was collaboratively integrated with the performer's actions to co-create the live performance on set. The resulting experience was highly personalised and perceptible only to the individual audience member, reinforcing intimacy and immersion within the theatrical encounter.





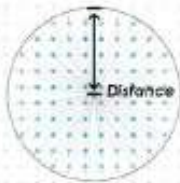
Immersive performance through audience-artist interaction



Workflow of blur for privacy between Grasshopper & Unity3D

STEP 1: Sight Distance

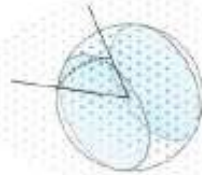
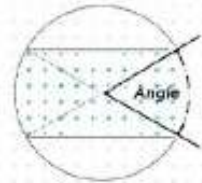
Top



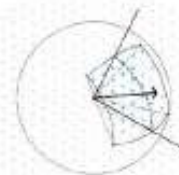
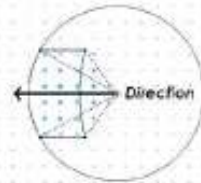
Parallel



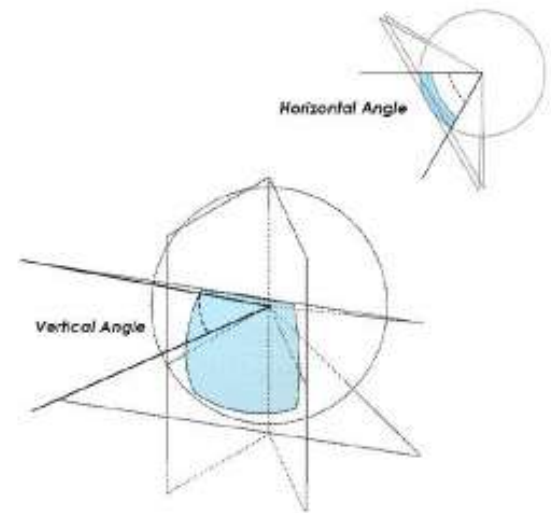
STEP 2: Horizontal View



STEP 3: Vertical view

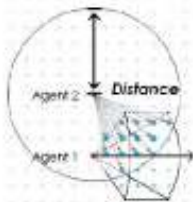


Horizontal Angle

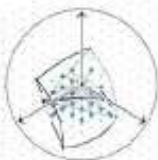


STEP 1: Sight Distance

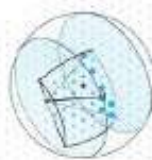
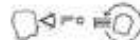
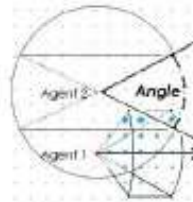
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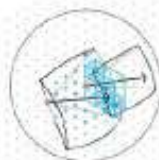
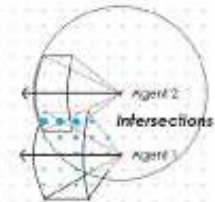
Parallel



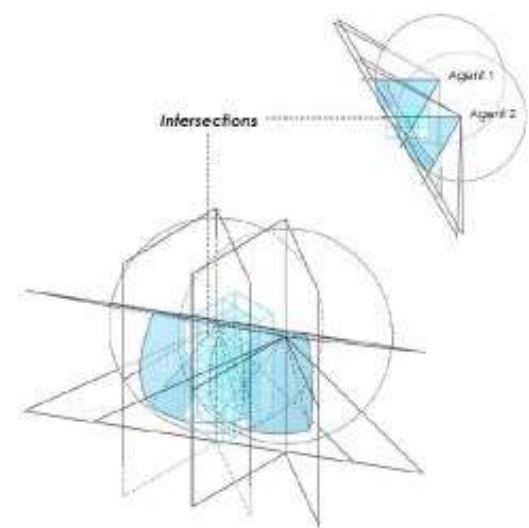
STEP 2: Horizontal View



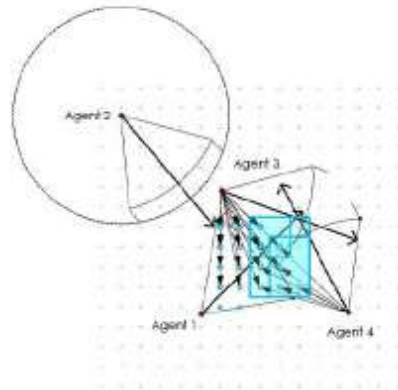
STEP 3: Vertical view



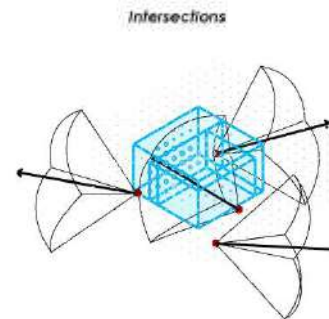
Intersections



Geometric exploration of angle of vision, to impose *blur*

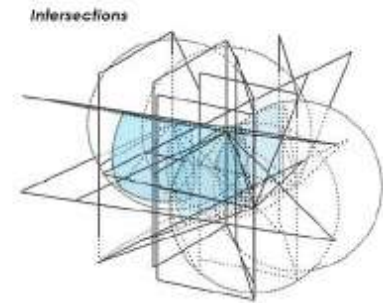
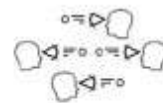


Top

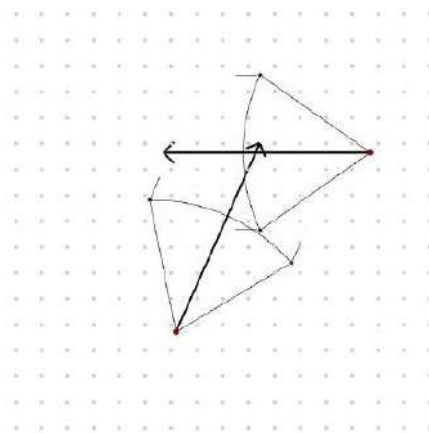


Intersections

Parallel



Intersections



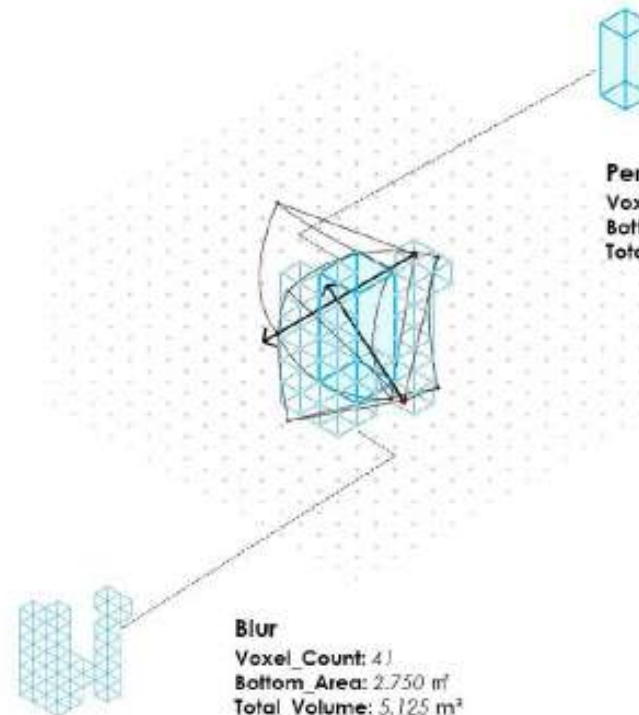
Field_voxel_size: 0.500m * 0.500m * 0.500m

Agent_1

Sight: 3m
Rotation_WorldXY_WorldXZ: (66, 0)
Position: (3.091, 1.857, 1.6)

Agent_2

Sight: 3m
Rotation_WorldXY_WorldXZ: (180, 0)
Position: (6.852, 5.327, 1.33)

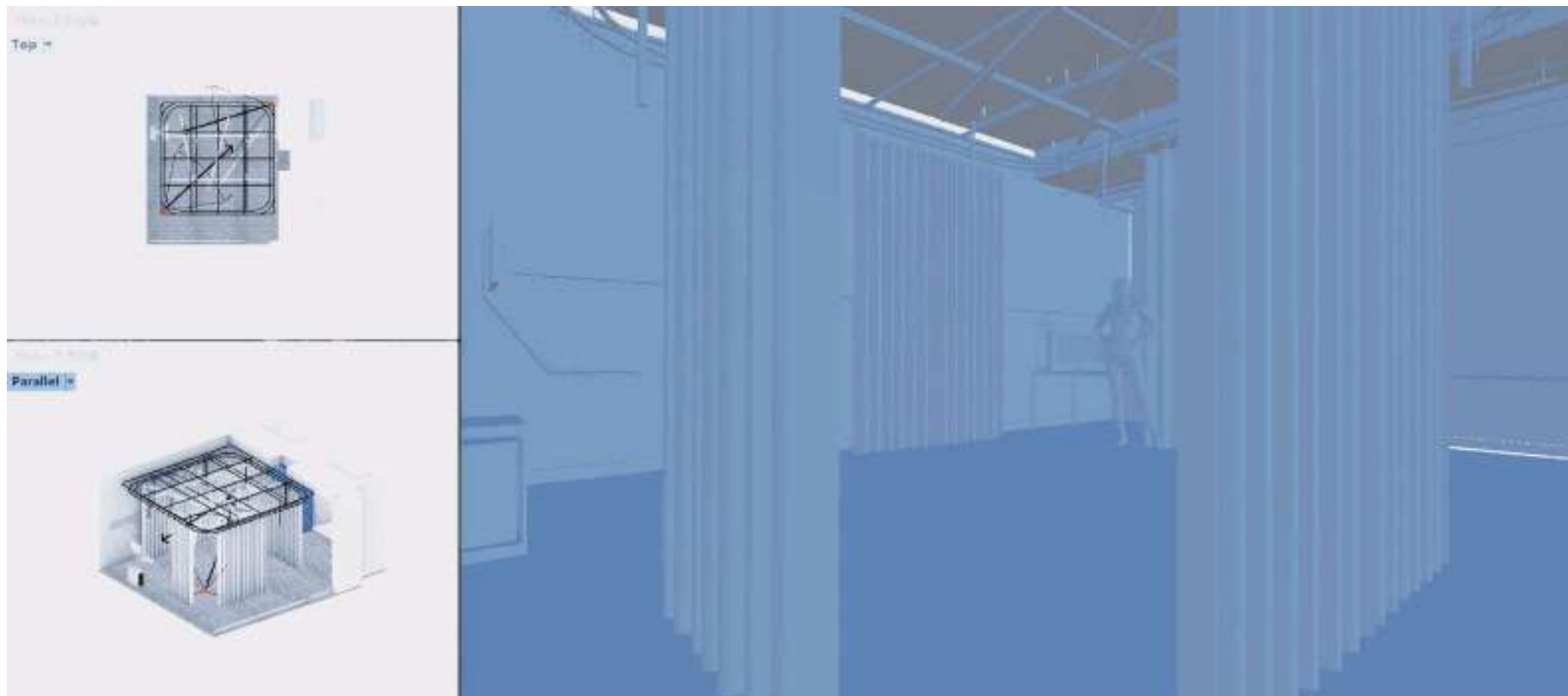


Performance

Voxel_Count: 8
Bottom_Area: 0.500 m²
Total_Volume: 2.500 m³

Blur

Voxel_Count: 41
Bottom_Area: 2.750 m²
Total_Volume: 5.125 m³



Field_voxel_size: $0.500m * 0.500m * 0.500m$

Agent_1

Sight: 3m

Rotation_WorldXY_WorldXZ: $(\alpha, 0) \alpha \in [0, 90]$

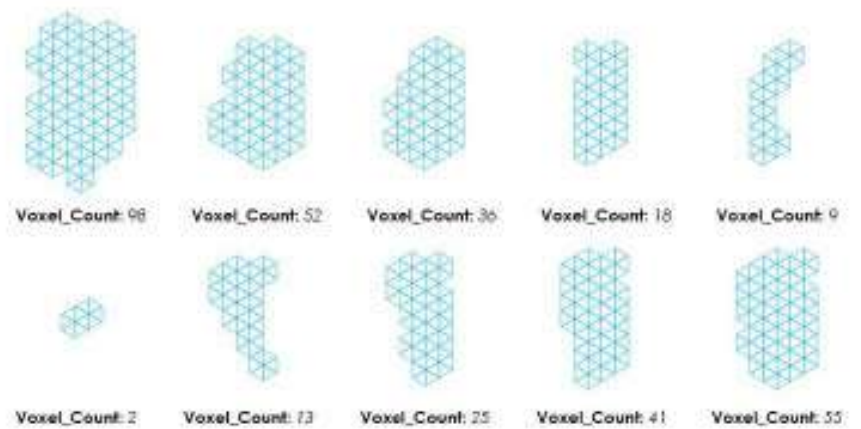
Position: $[1.211, 1.910, 1.600]$

Agent_2

Sight: 3m

Rotation_WorldXY_WorldXZ: $(0, 0)$

Position: $[4.775, 4.445, 1.330]$





Use WebCam to capture



Motion detection ML model



Generate landmark coordinates string

```
[150.7388949314224, 300.995946580244, -74.75793170928955,
144.14914463045218, 322.04119776772015, -74.81659282483777,
172.40434859025024, 320.742693114973, -74.78891203489573,
179.86591458320418, 319.65344694895337, -74.78843519973785,
148.87817788134084, 324.1289443759918, -72.19449281492585,
140.834040429924, 334.43970918455394, -72.08709714794875,
54.889522926807404, 106.06462340219114, -27.25246548682449]
```

UDP TRANSMISSION



Sender

Request
Response
Response
Response



Receiver

Split string into separate coordinates for each landmark

Point1.x	Point1.y	Point1.z
192.27753281593323,	275.39955854415894,	-203.3554937362671

Point2.x	Point2.y	Point2.z
201.51838668248173,	296.1868154438819,	-159.76145829067993

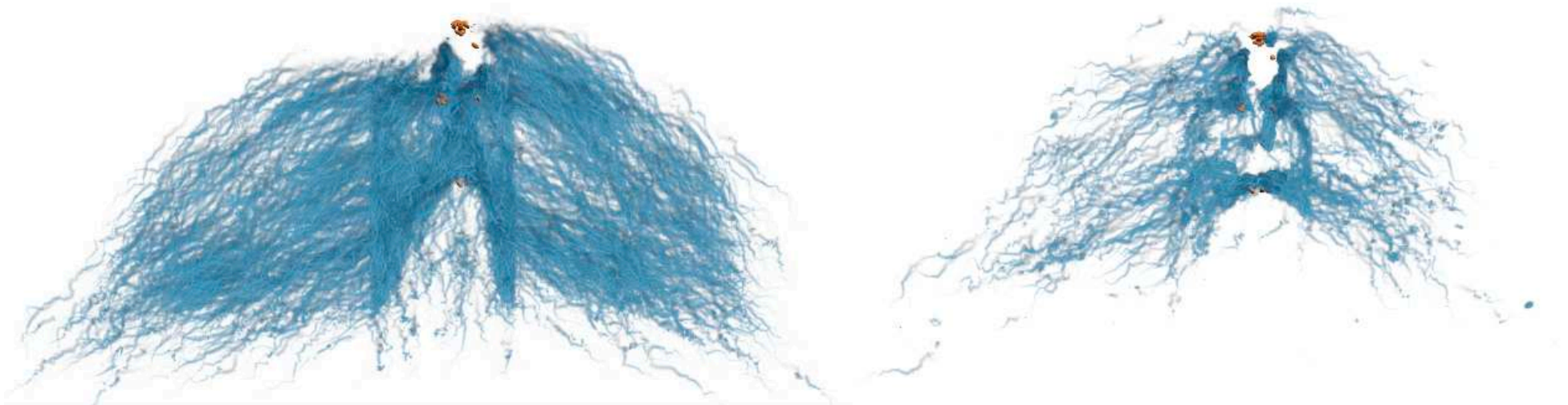
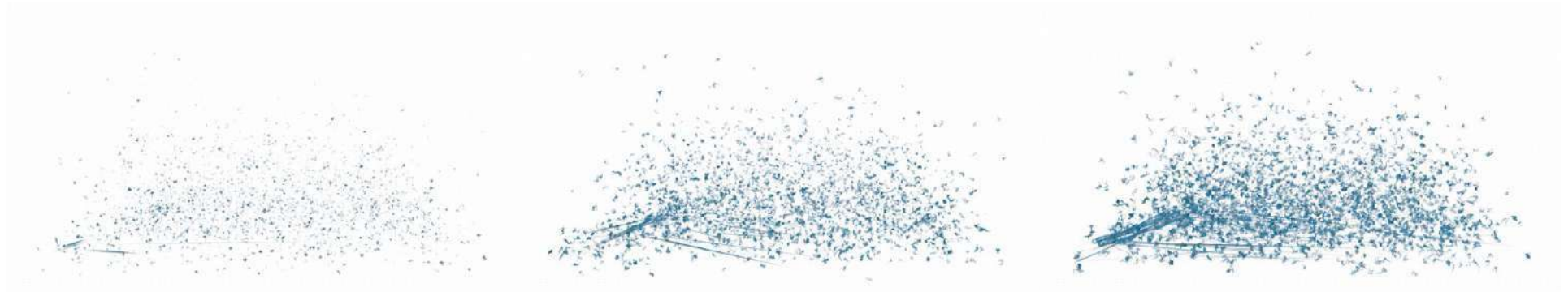
Build skeleton



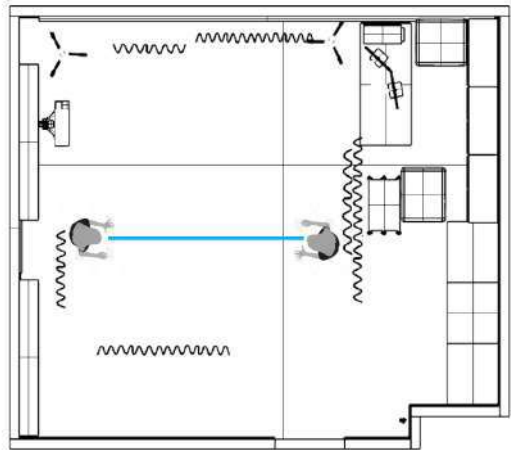
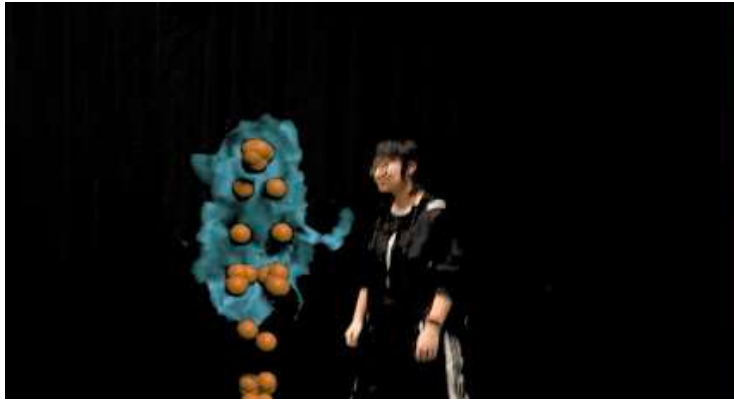
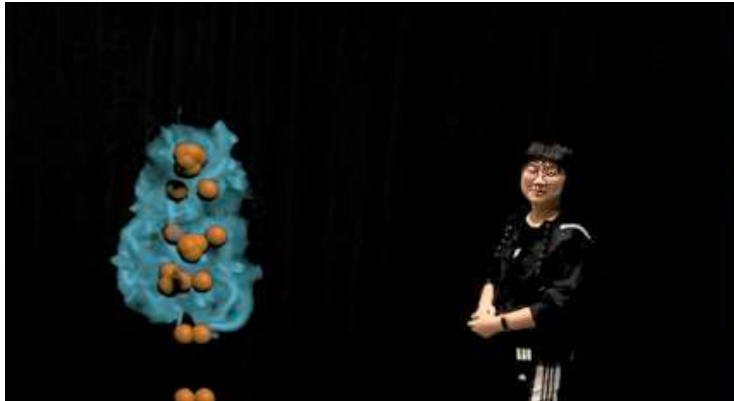
Assign the coordinates to the skeleton's joint



Workflow of Media Pipe Pose to Unity3D to detect articulation points



Agent-Based Modelling algorithm to formulate mesh along with attractor points



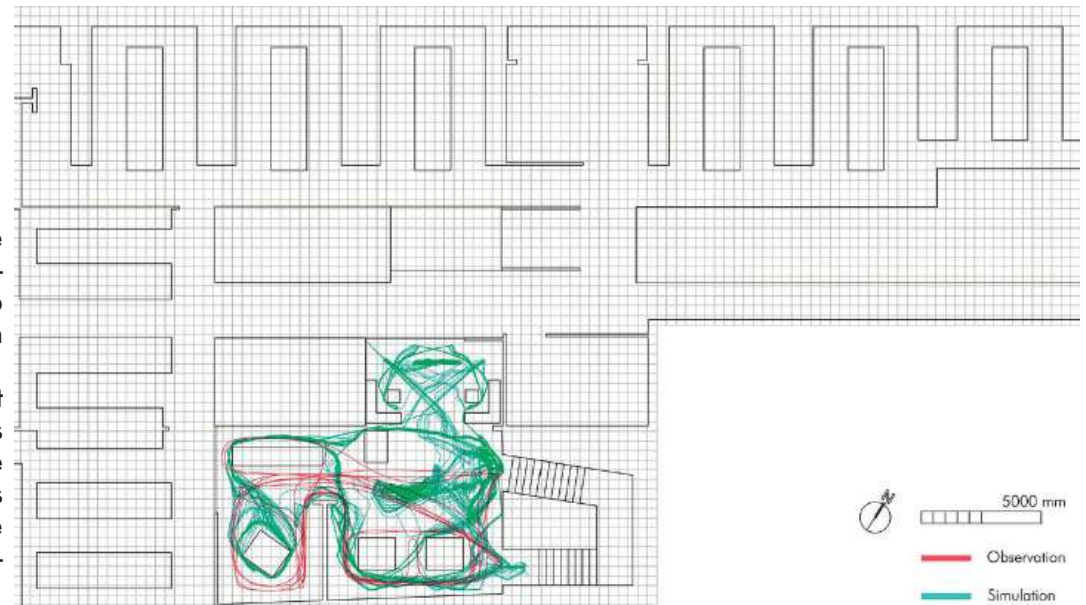
ISOVIST against OBSERVATION INSIDE an INDOOR EXHIBITION

MSc Thesis Project

Duration: 3 months - 2023

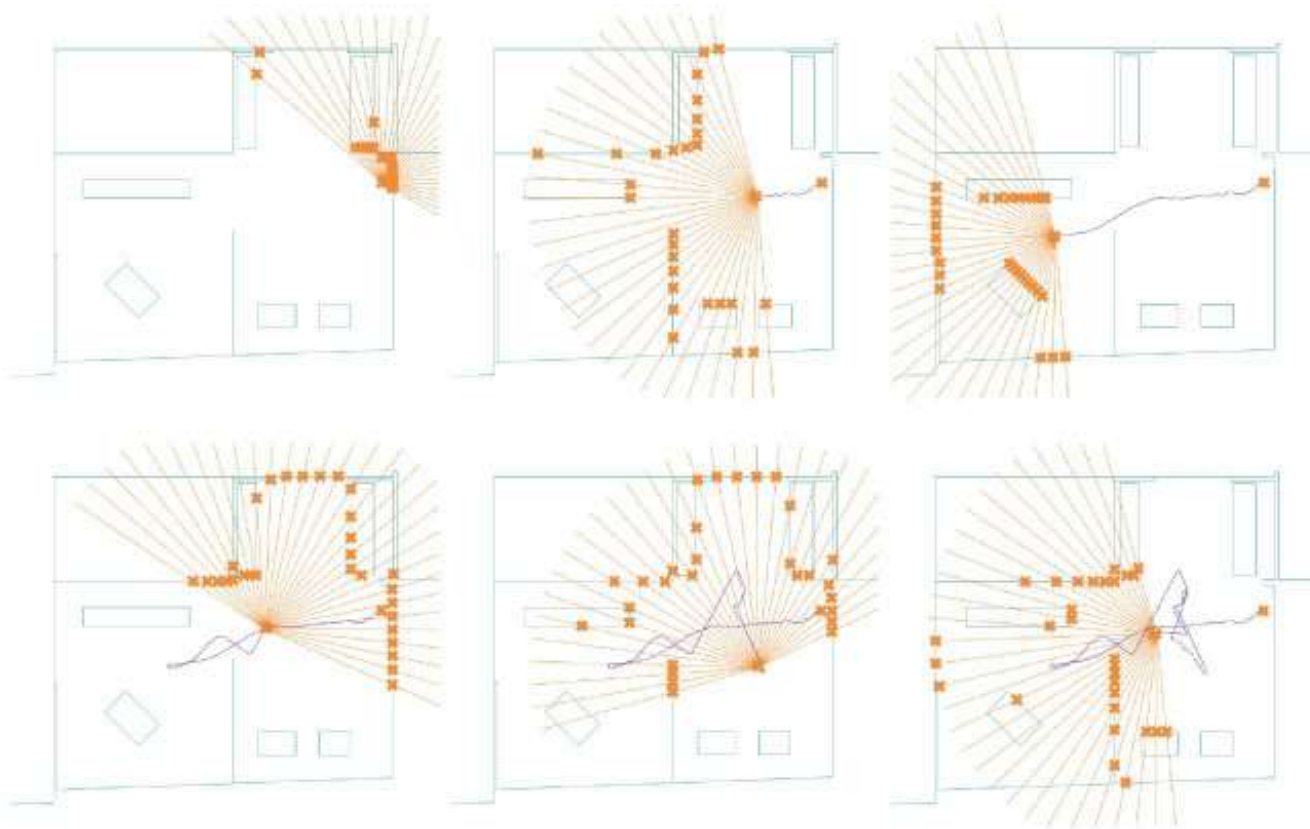
Tools: Rhino3D, Grasshopper, C#, Photoshop, Illustrator

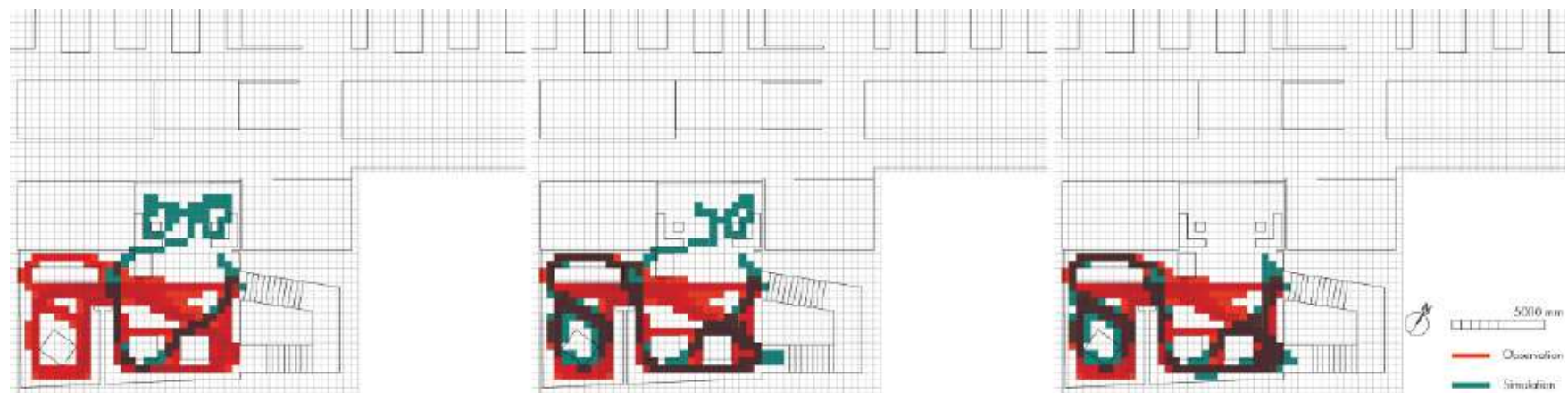
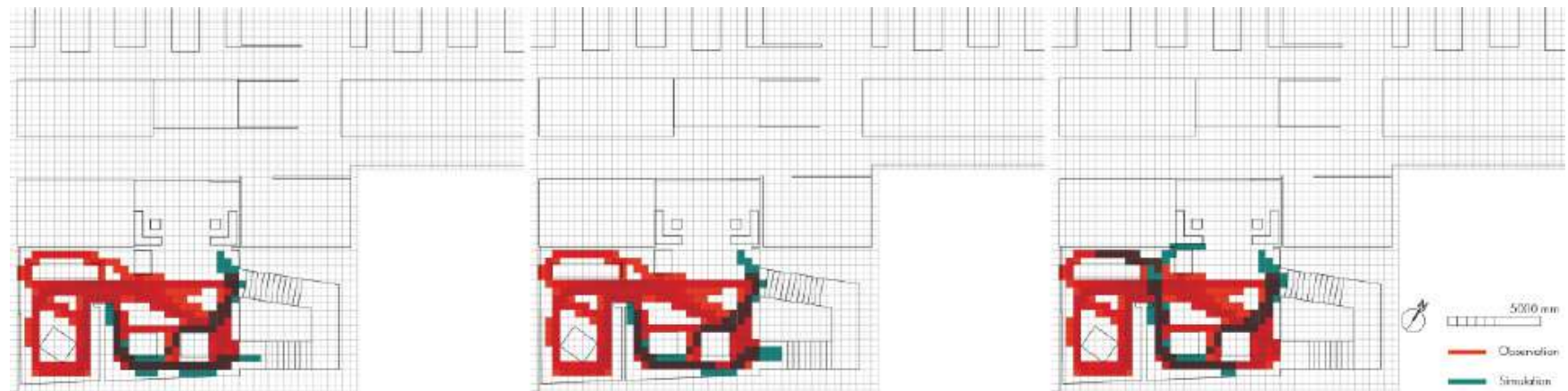
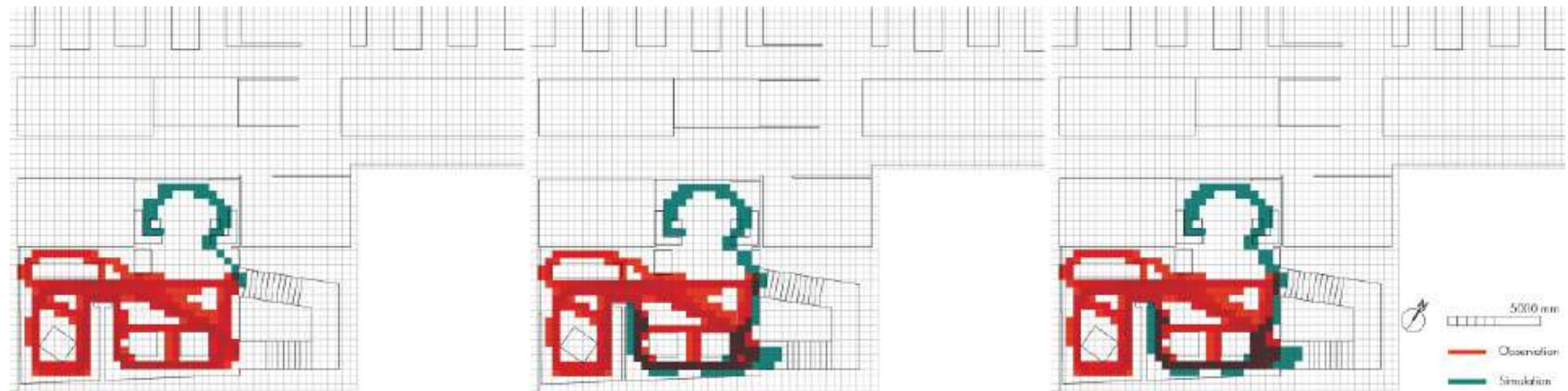
Although navigation in an indoor context is a widespread research interest, the research gap lies in combining real-life observation and computational simulation as methodologies. This research aims to contribute to such a research gap along with essential topics like indoor emergency evacuation that considers both the pedestrians with and without the knowledge of the context. To achieve such a result, 'movement trace' of space syntax theories, as the qualitative and isovist simulation as quantitative methods, has been used in this research. The findings evidence that the presence of destination points or goals impacts the difference between these two methodologies. Furthermore, the size of the context, the lines of sight or distance viewed, the number of hidden lines from the position of the pedestrians, the number of pedestrians and the number of obstacles to be avoided also influence such differences in the comparison.





Context & Isovist simulation





8 TOWNS RED

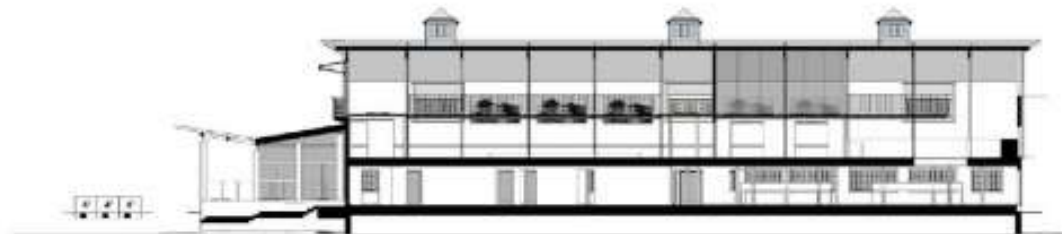
Professional Project - LineSync Architecture

Massachusetts, USA

Duration: 2018

Tools: SketchUP, Layout

This project was undertaken during my internship and involved the renovation of a 100-year-old barn house. The client was open to experimental ideas, which allowed us to explore a range of potential functions for the existing spaces. The site was backed by a mountain range, and I proposed the idea of creating a virtual enclosure that would function as a courtyard, framed by the existing massing of the building and the landscape beyond. This approach sought to establish a spatial dialogue between the architecture and its natural context. Within the interior, I recommended the introduction of voids to enable visual and interactive connections between different levels and spaces, enhancing spatial continuity and user engagement.





Proposed Renovation



Existing Site



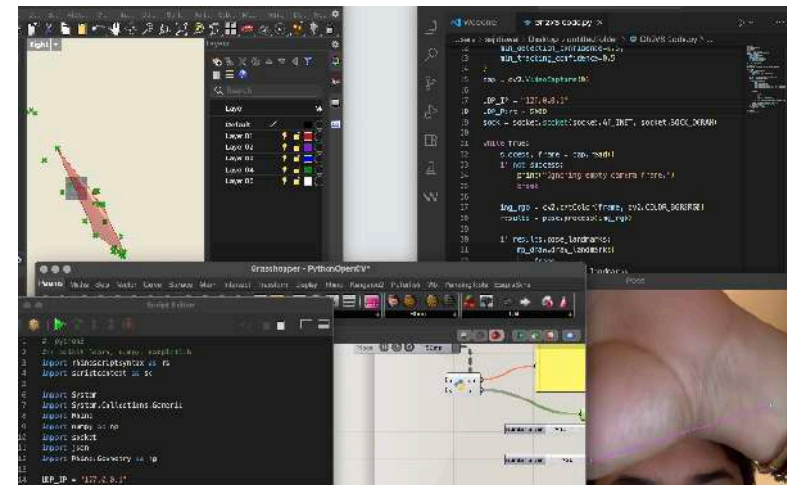
MediaPipePose ML in OPENCV to GRASSHOPPER

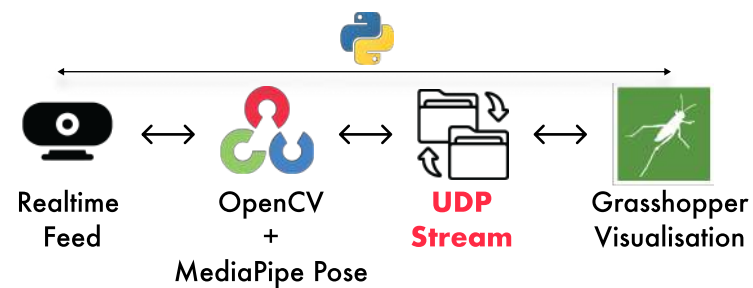
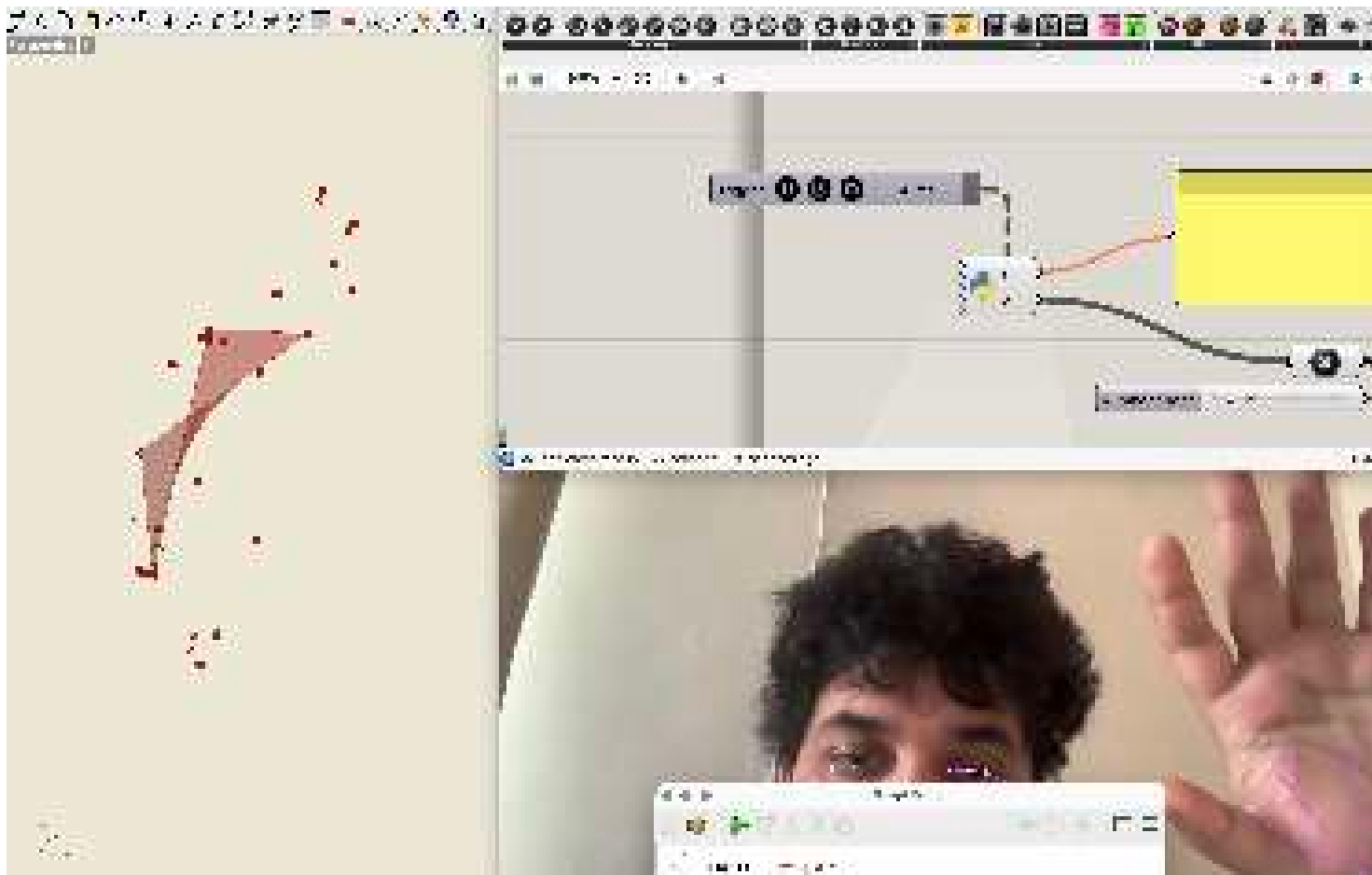
Experimental Project

Duration: 2/3 days - 2025

Tools: Python, OpenCV, Grasshopper, VS Code

This project explores real-time body-pose tracking by streaming MediaPipe Pose data from a Python–OpenCV environment into Grasshopper3D for spatial interpretation and prototyping. OpenCV and MediaPipe are used externally to detect and extract human articulation points, which are then transmitted as landmark coordinates via a UDP socket and reconstructed as points inside Grasshopper. A key limitation of the workflow is that Grasshopper cannot natively run or reliably interface with OpenCV, due to its constrained Python environment and dependency conflicts; therefore, a separate Python process and UDP communication were required as an intermediary. This also introduced practical drawbacks, such as manual configuration of IP addresses and ports, potential data loss due to UDP's non-guaranteed delivery, and limited synchronisation control. Despite these constraints, the setup demonstrates a viable bridge between real-time computer vision and architectural design tools, highlighting both the potential and current technical frictions of cross-platform, live data integration.





Workflow pipeline between Mediapipepose ML to Grasshopper

RAYCASTING3D In GRASSHOPPER

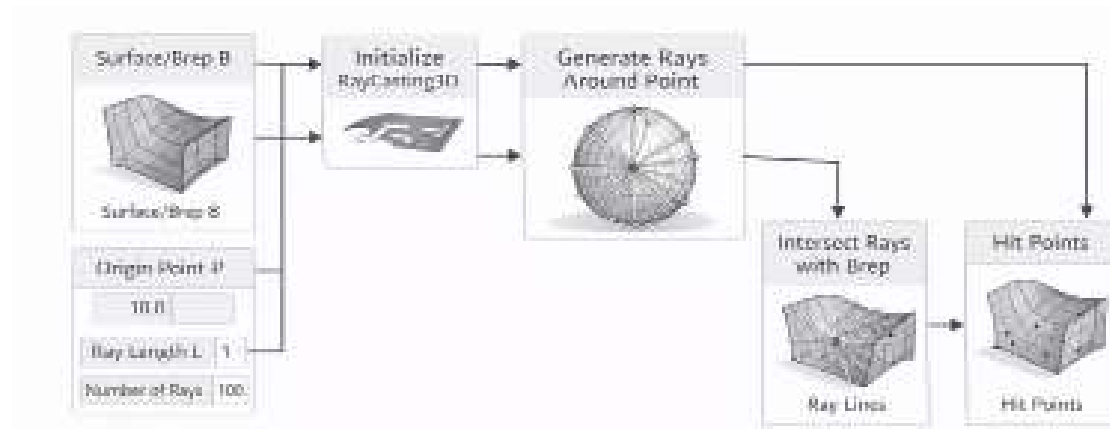
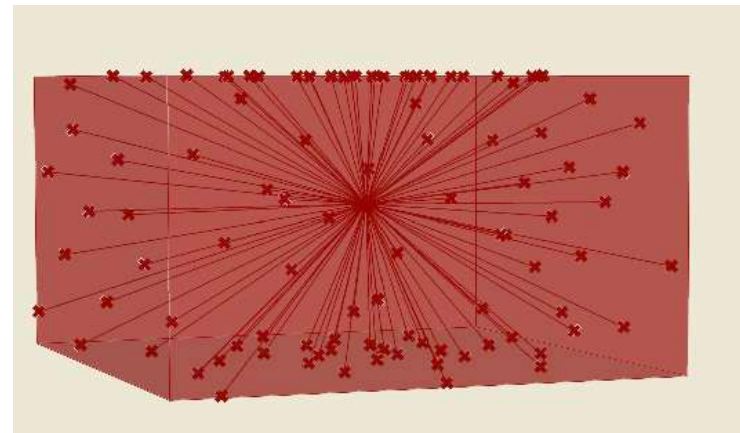
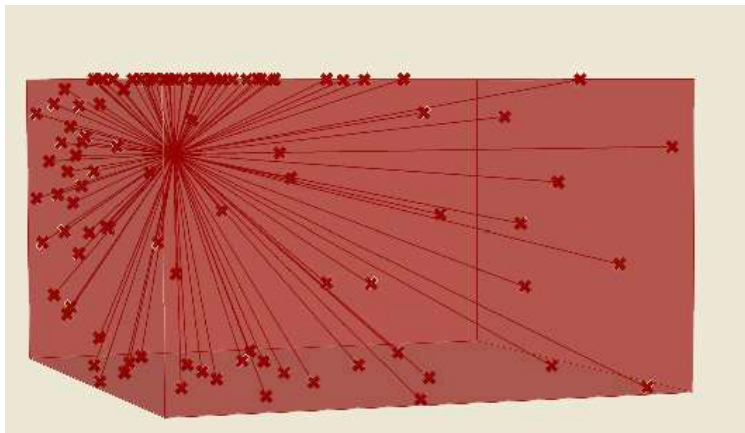
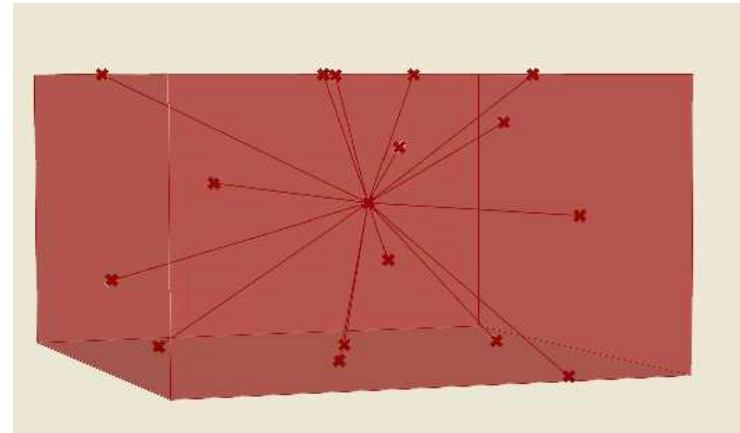
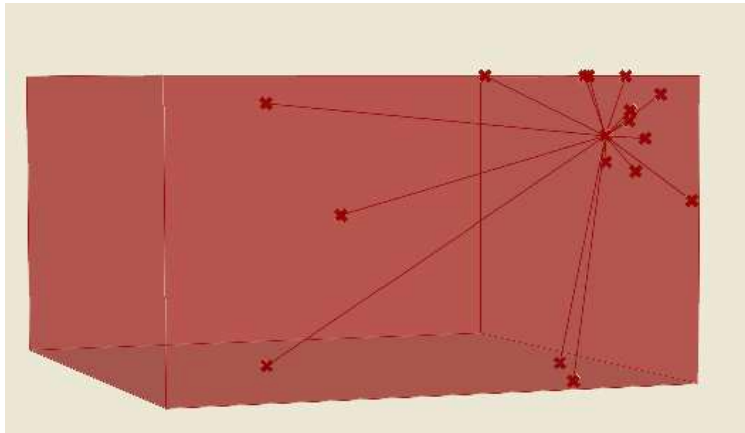
Experimental Project

Duration: 2 days - 2025

Tools: Python, OpenCV, Grasshopper, VS Code

This Grasshopper C# script performs 3D ray casting from a point towards a surrounding Brep to detect spatial intersections. Given a Brep (geometry), an origin point, a ray length, and a ray count, the script generates a set of rays distributed evenly over a sphere using a Fibonacci (golden-angle) sampling method, ensuring uniform directional coverage. Each ray is tested for intersection with the Brep, and when multiple intersections occur, only the closest hit point to the origin is retained. The output consists of two datasets: the full set of ray lines (trimmed to the intersection point if a hit occurs) and the corresponding intersection points. This method is useful for analysing spatial enclosure, visibility, proximity, or environmental interaction between a point and surrounding geometry in three dimensions.





Workflow of Raycasting3D algorithm



to be continued...